Daniel J Short Gianotti, Ph.D.

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Academic Appointments

2021 – 2026 **Research Scientist,** Massachusetts Institute of Technology.
Parsons Laboratory, Department of Civil & Environmental Engineering.

2016 – 2021 **Postdoctoral Associate,** Massachusetts Institute of Technology.
Parsons Laboratory, Department of Civil & Environmental Engineering.

Education

2011 – 2016 **Ph.D., Boston University,** Geography & Environment.
Thesis title: *The Potential Predictability of Precipitation over the Continental United States.*

1999 – 2003 **B.S., Harvey Mudd College,** Mathematics.
Thesis title: *Fluid Drop Coalescence in a Hele-Shaw Cell.*
Clinic title: *Internal Metrology for the JPL/NASA Space Interferometry Mission.*

Grants and Funding

Funded: 2024-2027
"Causes of Historical Change, Environmental Resiliency and Future of the Caspian Sea,"
Private Donor Grant.
PI: D Entekhabi, CO-Is: **DJ Short Gianotti**, MR Alizadeh. $494,747.

Funded: 2024-2025
"Global Mapping of Groundwater Recharge and Sustainable Aquifer Water Withdrawals Using Satellite Observations,"
MIT Climate and Sustainability Consortium.
First Author: MR Alizadeh, PI: D Entekhabi, CO-I: **DJ Short Gianotti**. $122,080.

Funded: 2024-2026
"Landcover Stability: Heating and Drying of Plants and Soils,"
MIT-Portugal Program.
First Author: **DJ Short Gianotti**, PI: D Entekhabi. $259,002.

Funded: 2020-2021
"Analysis of Water Relations in the Soil-Plant Continuum Using Microwave-Lidar Synergy,"
Fundació La Caixa & Massachusetts Institute of Technology,
PIs: D Chaparro & D Entekhabi, CO-Is: **DJ Short Gianotti**, AF Feldman, & T Jagdhuber.
€22,000, Grant Number 1673204776.

Publications and Talks

† denotes student advisee.
⋆ denotes presenting author.

Published Software Packages

Published Datasets

Professional Appointments

2021 – · · · · **Research Scientist,** Massachusetts Institute of Technology.
2016 – 2021 **Postdoctoral Associate,** Massachusetts Institute of Technology.
2011 – 2015 **Research Assistant,** Boston University.
2010 – 2011 **Math Teacher,** Boston Public Schools.
2004 – 2010 **Private Tutor**. .
2007 – 2008 **Lab Technician,** California Institute of Technology.
2005 – 2006 **Environmental Engineering Associate,** GV Jones & Associates.
2004 – 2005 **Research Assistant,** University of Alaska, Anchorage.
2003 – 2005 **Substitute Teacher,** Anchorage School District.
2004 **Staff,** National Youth Science Camp.
2001 – 2003 **Writing Consultant,** Harvey Mudd College.
2002 **Research Assistant,** Lawrence Berkeley National Lab.

Teaching

Teaching Fellow

2015 *Introduction to Quantitative Environmental Modeling* (Boston University)

Guest Lecturer

2016 – 2018 *Introduction to Hydrology and Water Resources* (MIT)
2017 *Introduction to Hydrologic Modeling* (MIT)

K-12 Instruction

2010 – 2011 *High school mathematics* (Boston Public Schools)
2004 – 2006 *All subjects, K – 12* (Substitute Teacher – Anchorage School District)

Private Tutoring

2002 – 2010 Math, physics, writing through advanced undergraduate
2006 – 2010 Chemistry, biology through introductory undergraduate
All subjects through advanced secondary

Mentorship

Thesis Committee Member

2022-2023 Meriah J Gannon (MIT MS in Climate, Environment, & Sustainability)
Thesis: *Propagation from meteorological drought to agricultural drought under climate change*

2021-2022 Lily N Zhang (MIT SB in Earth, Atmospheric, & Planetary Sciences)
Thesis: *Evaporative Controls on Convective Adjustment: a Satellite-Based Assessment of Convective Available Potential Energy (CAPE) During Surface Drydowns*

2018-2019 Apisada (Ju) Chulakadabba (MIT SB in Civil & Environmental Engineering)
Thesis: *Water and carbon flux responses to soil moisture pulses in the Western United States*

First-year Mini-UROP

2019 Naomi Lutz (MIT Civil & Environmental Engineering): *Impacts of Soil Moisture on Ecosystem Carbon and Water Exchanges*

Mentorship (continued)

- Nicole Toft (MIT Civil & Environmental Engineering): *Impacts of Soil Moisture on Ecosystem Carbon and Water Exchanges*

Undergraduate UROP

- Nicole Toft (MIT Civil & Environmental Engineering): *Land-atmosphere interactions at the inter-storm scale*

Professional Development

2015 *ComSciCon 2015* Communicating Science Workshop, Harvard University.

Professional Service

2022 *Climate Action Through Education (CATE) Workshop* K-12 Climate Curriculum Tuning Workshop, MIT Sloan School.

Conference and Workshop Organization

2025 *American Geophysical Union Fall Meeting* – Advances in Understanding Water-Energy-Carbon Interactions (B31A, B33J)
Primary Convener: Yanlan Liu, Conveners: **Daniel J Short Gianotti**, Flavio Lehner, Justin S Menkin, & Xiangtao Xu,

2024 *American Geophysical Union Fall Meeting* – Advances in Understanding Water-Energy-Carbon Interactions (B12A, B13A)
Primary Convener: Yanlan Liu, Conveners: Joshua Fisher, Xiangtao Xu, **Daniel J. Short Gianotti**, Justin S. Mankin, Yushu Xia, & Kevin P. Tu

2023 *American Geophysical Union Fall Meeting* – Advances in Understanding Water-Energy-Carbon Interactions (B43E, B52A)
Primary Convener: Yanlan Liu, Conveners: Xiangtao Xu, Vincent Humphrey, & **Daniel J Short Gianotti**

2022 *American Geophysical Union Fall Meeting* – Advances in Understanding Water-Energy-Carbon Interactions (B15B-I/II/III)
Primary Convener: **Daniel J Short Gianotti**, Conveners: Xiangtao Xu, Yanlan Liu, & Vincent Humphrey

2021 *American Geophysical Union Fall Meeting* – Advances in Understanding Water-Energy-Carbon Interactions (B010-I/II)
Primary Convener: **Daniel J Short Gianotti**, Conveners: Xiangtao Xu, Yanlan Liu, & Vincent Humphrey

Non-journal Reviews

2020 *Intergovernmental Panel on Climate Change* – Sixth Assessment Report
Second Order Draft for Working Group I (WGI)

2018 *Intergovernmental Panel on Climate Change* – Special Report on Climate Change, Desertification, Land Degradation, Sustainable Land Management, Food Security, and Greenhouse gas fluxes in Terrestrial Ecosystems
First Order Draft for US Global Change Research Program

Journal Reviews

- Remote Sensing of Environment
- Geophysical Research Letters

Professional Service (continued)

- Water Resources Research
- Bulletin of the American Meteorological Society
- Hydrology and Earth System Sciences
- Journal of Hydrometeorology
- Journal of Climate
- Biogeosciences
- Nature Communications

Memberships & Research Communities

- American Geophysical Union
- Boston Water Group
- Boston Area Hydrology Journal Club
- Harvard Plants & Climate
- IEEE

References

Available on Request

Updated October 27, 2026.
Find the most up-to-date version of this CV here.